

Convolution Neural Network in Precision Agriculture for Plant Image Recognition and Classification

Halimatu Sadiyah Abdullahi, Ray E. Sheriff, Fatima Mahieddine

School of Engineering and Computer Science.

University of Bradford, Bradford,

BD7 1DP UK

H.S.Abdullahi1@bradford.ac.uk, R.E.Sheriff@bradford.ac.uk, F.Mahieddine@bradford.ac.uk

Abstract— Agriculture is essential to the continued existence of human life as they directly depend on it for the production of food. The exponential rise in population calls for a rapid increase in food with the application of technology to reduce the laborious work, maximize production and reduce the impact of pressure on the environment. Precision Agriculture is thought to be solution required to achieve the production rate required. There has been a significant improvement in the area of image processing and data processing which has being a major challenge previously in the practice of precision Agriculture. A database of images is collected using remote sensing techniques and the images are analyzed to develop a model to determine the right treatment plans for different crop types and different regions. Features of images from vegetation's need to be extracted, classified, segmented and finally fed into the model. Different techniques have been applied to the processes from the use of the neural network, support vector machine, fuzzy logic approach and recently, the most effective approach generating fast and excellent results using the deep learning approach of convolution neural network for image classifications. Deep Convolution neural network is used in plant images recognition and classification to optimize production on a maize plantation. The experimental results on the developed model yielded results with an average accuracy of 99.58%.

Index terms— Convolution, Feature extraction, Image analysis, Validation and Precision Agriculture.

I. INTRODUCTION

Food is an essential necessity of human life and requires its continuous supply and production to cater the needs of the growing population through sustainable agriculture. This can be achieved with the application of emerging technologies in the sector to maximize production across a vegetation. Technology can aid/improve agriculture in several ways through pre-planning and post-harvest by the use of computer vision technology through image processing to determine the soil nutrient composition, right amount, right time, right place application of farm input resources like fertilizers, herbicides, water, weed detection, early detection of pest and diseases etc.

To achieve sustainable agriculture, Precision agriculture (PA) is used and it is the technology that enhances farming techniques. It prepares the land before planting, ensures an equally fertile vegetation across the field, monitors plantations during in-season growth, detects of early onset of pest and diseases, right amount, right time and right application of farm input resources to the right location through harvest and post-harvest processes [1][2]. PA involves remotes sensing, the use of Geographical Information System (GIS), Global Positioning

System (GPS) and data analysis. This technology enables the collection of field data in a known- destructive way and making the data available for analysis and implementation on the field [3]. The technology is made necessary due to a spatial and temporal variability of the field revealing information as, patterns and spatial relationships. The stages of remote sensing involve energy interaction with the atmosphere and interaction with the target, then the interaction of the energy with the sensor (camera), data transmission and processing, image analysis and finally the application of results to treatment areas required [4][5].

Simple applications on the farm involve determining the location of sampling sites, plotting maps for use in the field, examining the distribution of soil types in relation to yields and productivity. Other applications take advantage of the analytical capabilities of GIS and RS software for vegetation classification to predict crop yield, environmental impacts, modeling of surface water drainage patterns, tracking animal migration patterns and other ranges of applications [6].

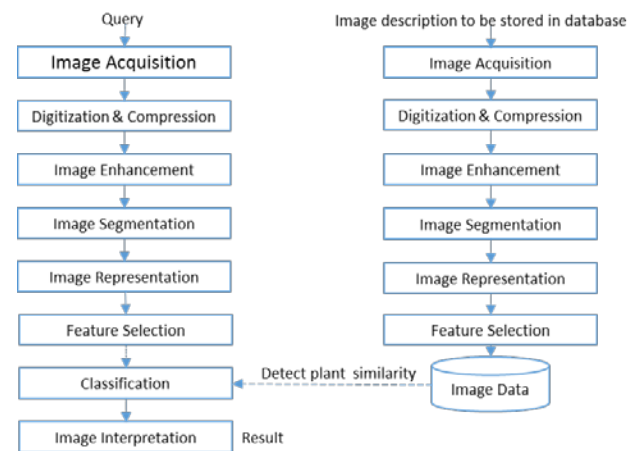


Fig. 1 Stages of Image analysis

PA techniques are employed to increase yield, reduce production costs, and pressure on the environment. With the potential of GIS analytical capabilities, variable parameters on a plantation that can affect agricultural production can be analyzed. These parameters that cause variability on the field are the composition of the soil, soil nutrients, soil water, soil physical properties, seed types used on the field, nutrients and herbicides application etc. [8][9].

Images obtained through remote sensing technology have advanced from the use of expensive resources (aircraft, satellites) and periodic availability of images to obtaining real time images with Unmanned Aerial Vehicles (UAVs), smart phones etc. at more affordable costs and in real or near real times [10]. Analyzing these images give information that can influence management decision on the farm for immediate treatment on from pre-planting to post-harvest stages. Information obtained from a vegetation can be derived from its shape, height, texture, color, growth rate to develop a pattern to model a farm land. Research into image analysis for Agricultural use has been active for over a decade now with several methods of data analysis. Figure 1 simplifies the key stages involved in the image analysis process.

In image acquisition, images are obtained using satellites, mobile phone, and the use of UAVs. A wide variety of imagery is available from satellites using both active and passive sensors which both operate from the microwave to the ultraviolet regions of the electromagnetic wave spectrum [7]. All of which vary in their spatial, spectral, radiometric and temporal resolution, playing an important role by identifying which applications the sensors are best suited for [11],[12]. With the satellites, the images obtained are pre-processed, matched together and can provide information about crop identification, crop area determination, and the crop condition. The challenge of this method of data acquisition is the associated cost of obtaining the data, the images are obtained only during visit cycles limiting its availability on demand for real time applications, and also, its poor resolution considering its long distance to the ground. The use of a mobile phone has the ability to produce excellent images but is limited in its ability to cover the entire field within a specific period set out.

Hence, the use of Unmanned Aerial Vehicles (UAVs) is considered as the best and most efficient of data collection on the field. The use of UAVs in agriculture is fast becoming widespread, while the implementation of aerospace engineering and sensor technology are reducing in cost. UAVs employ cameras to collect images and sensors to compile a set of data to help with monitoring and decision-making on-farm [13],[14]. UAVs gather information at very high resolutions which allow the differences in plantations to be easily noticed in centimetres as seen with the naked eyes [15]. They also provide immediate visual information about large areas of crops, which help farmers with fast decision-making.

The images obtained can then be imported into a GIS database and then stitched together using a software. Image enhancement involves the process of improving the image quality before processing; this involves noise removal, false color removal or detection, edge detection which can be done with simple image analysis technique in Matlab with fuzzy logic [16],[17],[18].

Before now, remote sensing through precision agriculture has been achieved in two step procedures, consisting of feature extraction and classification. The challenges in plant image recognition are due to the fact that all plants are very similar in shape and color representation and can easily be mixed up. Hence, the need to employ complex image processing to find high-level features of the plants for its classification.

II. BACKGROUND AND RELATED WORK

Image segmentation is a preprocessing stage that involves converting the image into segments based on uniformity. This is an important step necessary for efficient feature extraction. A variety of algorithms is usually developed to generate image objects. Image segmentation methods can be thought to be categorized in two main domains: i) knowledge driven methods and ii) data driven methods (bottom-up). In the knowledge driven approach, a model of the desired objects is defined, and then the extraction is performed while in bottom-up approaches, the segments are generated based on statistical methods [19]. In feature extraction, the attributes of the result from the image segmentation is used. The features are the shape, color feature, texture. Feature extraction is the process of parametrizing the farm images for defining efficient descriptors: texture, color or shape. Research in Precision Agriculture focuses on this two approaches; feature extraction and classification including anthropometric models, statistical methods, and histograms of gradients. The most successful methods used are; artificial neural network and support vector machines with both methods combined together to achieve better feature extraction results [20].

Artificial neural network (ANN) in machine learning is an information processing systems and intelligent programs inspired by biological neural networks consisting of the computational system of structure, processing method and learning ability. ANN is characterized by a large number of very simple processing neuron-like processing elements, a large number of weighted connections between elements, distributed representation of knowledge over the connections. Knowledge is acquired by the network through the learning process. Different network architectures require different learning algorithms. There is the supervised learning where the network is provided with the correct output for every input pattern. The weights are determined to allow the network produce results as closely as possible to the known or set output results. The back propagation belongs to this category for checking errors back in the process [21].

The unsupervised learning does not require a known output associated with the input pattern in the training set. It explores the underlying structure of the data set or similarity between patterns and organizes data into categories from these correlations. The other category is the hybrid comprising of both the supervised and unsupervised learning with part of the weights determined through both processes. Based on architecture, there are two main types of ANNs; feed-forward and feed-backward. Feed-forward ANNs are where the output of any layer is not likely to influence the same layer and feedback consists of the signals travel in both directions by involving loops in the neural network [22]. The simple model of ANN is shown in figure 2 below:

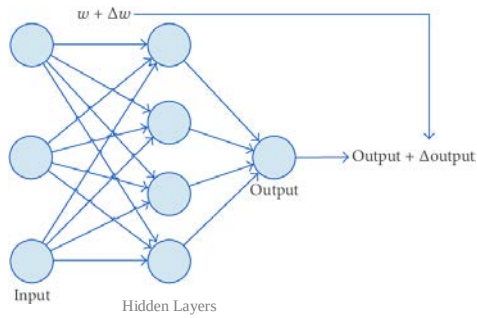


Figure 2: Simple Schematic model of an Artificial Neuron Network

In this paper, a new approach is used for classifying and recognizing the health state of the plantation and immediately generating treatment solutions on the go.

There are two contributions in this paper; convolution Neural Network (CNN) is used to classify and recognize different classes of the plant images, detect plant diseases and determine the rate of growth of plantations at the same time respectively by extracting all features during the analysis and these features are automatically learned to facilitate the development of systems that learn from one end to the other. Also, the results are visualized to enable generation of treatment plans for immediate application on the plantations and to also provide insight on how the models perceive the leaves on the images. The novelty of this approach is in its simplicity and ability to detect different features during analysis accurately and efficiently: both the healthy leaves and its background images are aligned with other classes which effectively enables the model to distinguish healthy from unhealthy plants while mapping the problem areas on the farm.

With precision Agriculture, implementation of management decisions on the farm like the right application of resources; fertilizer, herbicide, water or nutrients for preparing the soil to make it fertile can be made easily and at the right time, quantity and location with the results from the image analysis. This will lead to maximizing production and informing on crop health and disease detection. For an automated farm operation, the results of the processed images obtained from the sensors are immediately sent to the farm machinery or unmanned ground vehicles with prescribed amount of treatment for immediate application on the farm. There have been other image processing techniques used to achieve this aim, CNN outperforms these techniques by presenting more accurate results efficiently. Authors have reviewed, presented and recognized the growing need to develop sensors and image analysis to suit these purposes while implementing various technologies to increase agricultural production and maintaining the environment at the same time.

Hulya Yalcin et al proposed a CNN architecture to classify the different types of plants from image sequences. The results obtained was compared with others from using SVM classifier with different kernels as well as feature descriptors of LBP and GIST and the performance of CNN achieved an accuracy of 97.47% which outperforms the other methods with an accuracy between 74.92 to 89.94% [24].

Also, T. Rumpf et al developed a procedure for detecting and differentiating diseases on sugar beets using similar technique using support vector machine and vegetation spectral

indices. The discrimination between healthy sugar beet leaves resulted in classification accuracy of 97% with the multiple classifications between healthy leaves and leaves with symptoms of three diseases achieved an average accuracy of 86% but the results did not achieve the desired level of accuracy and not in real time [26].

In the report by Y. Lanthier et al, a comparative study was carried out between both the supervised pixel oriented and the object oriented classification based on image segmentation in precision agriculture using hyperspectral images. Images were acquired using the CASI sensor and statistical comparison is used to determine the mean difference to neighbor objects confirmed that the segments had minimum mixing effects in respect to other segmentation levels and neighboring ground entities [27]. The results achieved were fast and accurate specific for certain crops but not capable of detecting multiple features in the processing. Recently, some authors have also used neural network and SVM to diagnose plant diseases using image processing by training a network to develop a model to identify plant leaves. The images of the leaves are converted to RGB to Hue Intensity Saturation or lab color space and the leaf disease segmentation is done by using hierarchical clustering [28].

Arivazhagan S et al went further to highlight the relevance of the automatic detection of plant diseases using image analysis by proposing an algorithm with SVM classifier that successfully detected and classified plant diseases with an accuracy of 94%. The experiment was performed with a database of 500 plant leaves with 30 different native plant species confirming the robustness of the approach. Application of Texture analysis was used in detecting and classifying the plant diseases [29]. The first research implementing the use of CNN approach with great results was performed by Srdjan Sladojevic. In his model, CNN was successfully used to classify and identify plant diseases with results achieved having a precision of between 91 – 98% for separate class tests and an average performance of 96.3% on 13 different types of plant diseases out of healthy leaves with the ability to differentiate the leaves from their surroundings [25]. The model was based on a single classifier developed to only detect plant diseases for the plantation. CNN was also used by (Krizhevsku, 2012) to achieve a top 5 error of 16.4% for image classification with a thousand categories of data classes [23]. The error rate has since reduced significantly with the availability of the large dataset of images (over 10,000) collected by the plant village made freely available on the internet.

III. METHODOLOGY

Using Convolution neural network (CNN), also known as ConvNet, a network can be trained from scratch with a large data set or fine tuning an existing model or making use of “off the shelf Convolution neural network features”. Fine tuning involves transferring weights of the first ‘n’ layers learned from a previous base network to the new network. The dataset obtained for the new network is now trained to perform specific tasks. CNN can efficiently learn generic image features and the features can be used with simple classifiers to solve most computer vision challenges. The process involves taking off the last layer of the trained CNN and using the activations

of the last connected layer as features; it is used in this stage as a feature extractor instead of a classifier. Research has shown that this approach can be used for a dataset with a small number of images which was a challenge previously in producing accurate results. It is also reported to outperform both the fine tuning and training from the scratch approach by producing results with greater accuracy with both small and larger datasets [30]. The model architecture for a CNN is shown in Figure 2; the model is configured to take a fixed sized image of 224 x 224 RGB as an input for processing. All the training images are immediately center normalized and the CNN layers are made of 13 layers of the filters respectively. The model consists of the pooling layers and the fully connected layers (FC).

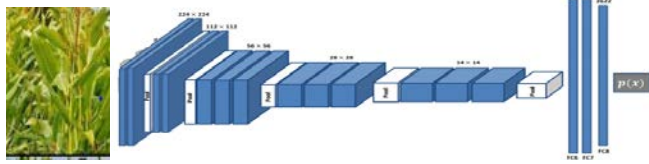


Figure 3(a): CNN model architecture

The first two convolution layers have 4096 channels, FC8 has 2622 channels used in the classification. The overall framework for the approach is represented in Figure 3(b).

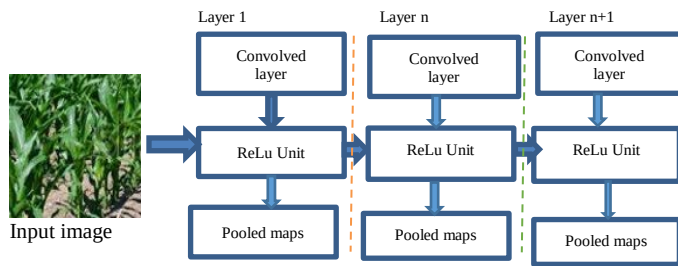


Fig 3 (b): Overall framework of CNN

From several research, CNN has been used for plant disease classification, fruit detection with transfer learning using imagery obtained from two modalities (RGB and NIR) information using late fusion technology to combine both information sources [31]. Deep CNN has also been used for automatic plant identification from image sequences collected from smart agro- stations [24] and has been used to perform weed detection and classification with UGV applied to the input RGB + Near infra-red images [32]. The UGV adjusts the application of the herbicide to its required use. CNN has been used in the classification of agricultural pest insects by computing a saliency map with the objective of an automated visual system to provide expert- level pest insect recognition with minimal operator training [33].

This technique is used in a novel way of training the images for quick system implementation in practice. Appropriate and properly labeled data are required for this process from training to evaluation stages, the images obtained were augmented to increase the data set and avoid overfitting during the training process. The images were pre-processed and labelled using the MS picture manager. A total of about 1918 pictures were obtained and was augmented to 4588 images for better results and training. A single PC was used for the entire process of training and testing to determine healthy, unhealthy and partially unhealthy plants with the CNN training performed on

the graphics processing unit. With convolution neural network, better results are obtained with larger datasets than only with the fewer number of images.

A. Image processing and labelling:

Images obtained from the farm of healthy, partially healthy and unhealthy plants were about 1900 and were converted into the same format (jpeg) and into a square matrix for greater resolution and quality during processing. Table 1 shows the total number of images obtained with the augmented images and figure 4 shows some of the images taken on the plantation and their classes.



Fig 4.1 (a) Unhealthy plants



Fig 4.2 (b) partially unhealthy plant



Fig 4.3 (c) Healthy Plants

TABLE I. IMAGES OBTAINED FOR THE COMPUTATION

Class	Number of Original Images	Total number of images (Original and Augmented)	Number of images used for validation/testing
Healthy plants	861	1221	122
Partially Healthy plants	479	1578	157
Unhealthy	578	1789	198

IV. EXPERIMENTS AND RESULTS

A pre-trained 16 layered convolutional neural network (the VGG16) [34] is used to extract features from our images. These deep representations (features) comprise of both low and high-level patterns. The low-level information includes plant color, height, smoothness and the sparsity of vegetation etc. while high-level features are more specific details such as the overall appearance of the plant. The learned filters are tasked with the responsibility of extracting all features represented in the plant leaf images.

Support vector machine (SVM) classifier was trained to recognize and identify an unknown/unseen plant image using two tests:

Test 1: Considering healthy (H) and unhealthy (U) images.

The extracted deep features were fed into SVM classifier to achieve binary classification. The partitioning protocol utilized is the 10 folds cross validation technique [35]; the data was divided into 10 partitions, 9 parts used for training the classifier, and then prediction was performed using the left out 10%. This procedure was repeated 10 times. The results obtained from the 10 folds were averaged to get the overall performance (success rate). The visualization of the image features in trained classification model is represented in figure 5 below: The result is presented with receiver operating characteristics (ROC) s that illustrates the performance of the classifier with its discrimination threshold varied. The ROC is used to represent the trade-offs between true positive and false positive rates of the classifier [36],[37].

The recognition/identification rate achieved in our first test was 99.58%. The receiver operating characteristic (ROC) is shown below.

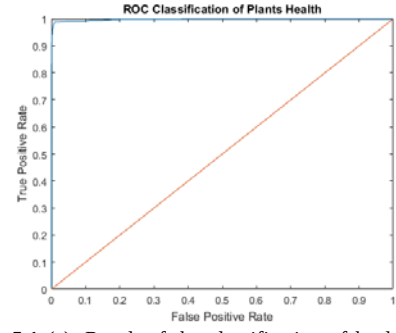


Fig 5.1 (a): Result of the classification of healthy and unhealthy maize plants

Test 2: Considering three types of plants Healthy, Partially Healthy (P), and Unhealthy: The binary classification, conducted in 1 above was repeated 3 times

a) H plants considered as class positive samples (+1), while P and U were labeled the negatives (-1)

Result: 88.3%

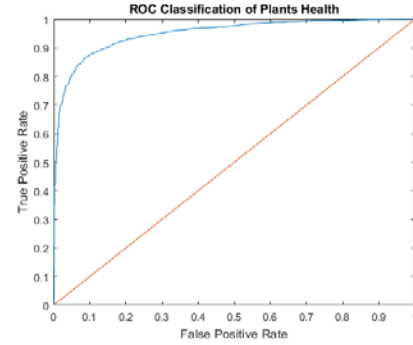


Fig 5.2 (b): P as labelled +1, U and H as -1

80.9%

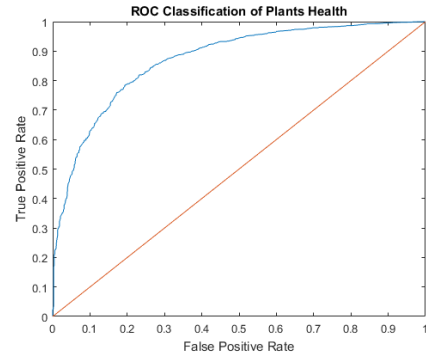


Fig 5.3 (c): U as +1, H and P considered as -1

96.5%

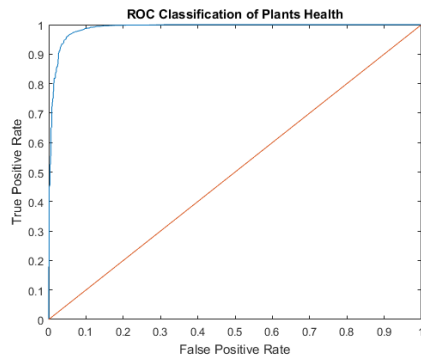


Fig 5.4(d)

Recognition or identification rate is computed as the percentage of images that were successfully identified as H, U or P.

V. DISCUSSION

Modeling a ConvNet for training requires millions of data, heavy computational power and complex mathematics which takes a long time to generate results [38] and transfer learning takes off the last layer of a trained network from the neural network. This technique is applied here by using ConvNet that saves what it learns in weights and fine tuning the learned weights from existing architecture. The last layer of an image is mapped on to the new image i.e. the existing ConvNet with the existing layer is used to train the database [39],[40],[30],[41]. For training and testing, K false cross validation is used and the database is split into two parts: one set for training and the other for validation. The 10 fold is usually the best with 9 sets of data used for training the algorithm and 1 set for testing. As mentioned earlier, VGG nets are used to extract features and the images are formed by a set of numbers (vectors) [42]. The red line graph depicts random guesses showing that the performance of the algorithm suggesting its excellence. The true positive shows how many images are classified as healthy, partially healthy and unhealthy in the three graphs while the confusion matrix shows the true positive and the true negative results. Most of the classifiers use several binary classifications and then combine them. The binary classification, conducted in the experiment above was repeated 3 times for:

- H plants considered as class positive samples (+1), while P and U were labeled the negatives (-1)
- P as labeled +1, U, and H as -1
- U as +1, H and P considered as -1

For the first experiment, the healthy plants are seen as positive while the partially healthy and unhealthy are seen as the negatives. The recognition rate was for the 88.3% when the H was considered as positives and both P and U considered as negative. The success rate is slightly lower with a lower number of images in the training set for the partially healthy plants. The best recognition rate and top success obtained was 99.58% making it the best result ever achieved for detecting classes of healthy from unhealthy plants. The recognition rate when P is labelled as positive and U and H are labeled as unhealthy yielded 80.9%. The performance of the algorithm suggests also that, the quality and resolution of data obtained plays a major role in the classification process. Clear and good labeled images produce higher accuracy than unclear and improperly labeled

images. Also, the larger the data set, the better the accuracy of classification for training with CNN.

With these excellent classification results, treatment plans can be easily formulated and stressed areas highlighted with treatments for farm management decisions in the shortest time possible. The results can be sent to auto steer tractors to apply the input resources or mapped out for treatment by the farmers with their GPS coordinates. Other factors to be considered in the developed model are; the region with different soil composition, weather, the soil type or quality, stages of pests and disease infection, soil nutrient availability, soil water requirements and the atmospheric conditions etc. to determine the required and specific treatments for farm output optimization.

VI. CONCLUSION

The use of off-the-shelf ConvNet representations for the problem of estimating plant health on a maize plantation was used with an average prediction accuracy of 99.58%. So far this is the best result achieved when compared with other techniques and proves the ability of the model to accurately predict the treatment solutions to produce an equally fertile land for optimization of production. This model can be developed for different regions with separate soil composition, weather and other factors from one geographical region to another for making informed decisions on a plantation.

REFERENCES

- 1] R. Bongiovanni and J. Lowenberg-Deboer, "Precision agriculture and sustainability," *Precis. Agric.*, vol. 5, no. 4, pp. 359–387, 2004.
- 2] J. V. Stafford, "Implementing Precision Agriculture in the 21st Century," *J. Agric. Eng. Res.*, vol. 76, no. 3, pp. 267–275, 2000.
- 3] A. McBratney, B. Whelan, T. Ancev, and J. Bouma, "Future directions of precision agriculture," *Precis. Agric.*, vol. 6, no. 1, pp. 7–23, 2005.
- 4] N. Zhang, M. Wang, and N. Wang, "Precision agriculture - A worldwide overview," *Comput. Electron. Agric.*, vol. 36, no. 2–3, pp. 113–132, 2002.
- 5] H. S. Abdullahi, F. Mahieddine, and R. E. Sheriff, "Wireless and Satellite Systems," vol. 154, no. January 2016, pp. 388–400, 2015.
- 6] E. S. Government of Alberta, Alberta Agriculture and Rural Development, Policy and Environment, "What Is Precision Farming?" Taber, Alberta, Canada, 1997.
- 7] T. Blaschke, "Object based image analysis for remote sensing," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 65, no. 1, pp. 2–16, 2010.
- 8] C. Brown, "GIS, GPS, and Remote Sensing Technologies in Extension Services Where to Start, What to Know," pp. 1–6, 2016.
- 9] T. Shi, "Ecological agriculture in China: Bridging the gap between rhetoric and practice of sustainability," *Ecol. Econ.*, vol. 42, no. 3, pp. 359–368, 2002.
- 10] C. Zhang and J. M. Kovacs, "The application of small unmanned aerial systems for precision agriculture: A review," *Precis. Agric.*, vol. 13, no. 6, pp. 693–712, 2012.
- 11] P. Mondal, M. Basu, and P. B. S. Bhadoria, "Critical Review of Precision Agriculture Technologies and Its Scope of Adoption in India," *Am. J. Exp. Agric.* 1(3) 49-68, 2011, vol. 1, no. 3, pp. 49–68, 2011.
- 12] Natural Resources Conservation Service, "Precision

- Agriculture: NRCS Support for Emerging Technologies,” 2007.
- [13] “UAVs and Precision Agriculture #15.” [Online]. Available: <http://aerialfarmer.blogspot.co.uk/2014/03/uavs-and-precision-agriculture-15.html>. [Accessed: 08-Mar-2015].
 - [14] D. Wright, V. Rasmussen, R. Ramsey, D. Baker, and J. Ellsworth, “Canopy Reflectance Estimation of Wheat Nitrogen Content for Grain Protein Management,” *GIScience Remote Sens.*, vol. 41, no. 4, pp. 287–300, 2004.
 - [15] K. J. Hayhurst, J. M. Maddalon, N. A. Neogi, and H. A. Verstynen, “Safety and Certification Considerations for Expanding the Use of UAS in Precision Agriculture,” *International Society of Precision Agriculture, 13th Annual Conference*. pp. 1–15, 2016.
 - [16] “UAV Cameras and Sensors, Precision Agriculture.” [Online]. Available: <http://www.thedroneinfo.com/2015/03/13/uav-cameras-sensors-precision-agriculture/>. [Accessed: 15-Mar-2015].
 - [17] J. Barnard, “Small UAV Command, Control and Communication Issues,” *with UAVs... with UAVs*, vol. 2010, no. 15 November 2010, pp. 75–85, 2007.
 - [18] F. G. Costa, J. Ueyama, T. Braun, G. Pessin, F. S. Osorio, and P. A. Vargas, “The use of unmanned aerial vehicles and wireless sensor network in agricultural applications,” *2012 IEEE Int. Geosci. Remote Sens. Symp.*, pp. 5045–5048, 2012.
 - [19] L. Wang, J. Shi, G. Song, and I. Shen, “Object Detection Combining Recognition and Segmentation,” *Lecture Notes in Computer Science*, vol. 4843, no. 2, p. 189, 2007.
 - [20] K. A. Mate G Pooja, Singh R Kavita, “Feature Extraction Algorithm for Estimation of Agriculture Acreage from Remote Sensing Images,” pp. 5–9, 2016.
 - [21] A. K. Jain, J. Mao, and K. M. Mohiuddin, “Artificial neural networks: A tutorial,” *IEEE*, vol. 29, no. 3, pp. 31–44, 1996.
 - [22] C. C. Yang, S. O. Prasher, J. A. Landry, H. S. Ramaswamy, and A. Ditommaso, “Application of artificial neural networks in image recognition and classification of crop and weeds,” *Can. Agric. Eng.*, vol. 42, no. 3, pp. 147–152, 2000.
 - [23] K. Alex, I. Sutskever, and G. E. Hinton, “Imagenet classification with deep convolutional neural networks,” in *Neural Information Processing Systems (NIPS)*, 2012, pp. 1097–1105.
 - [24] H. Yalcin and S. Razavi, “Plant Classification using Convolutional Neural Networks.” 2016.
 - [25] S. Sladojevic, M. Arsenovic, A. Anderla, D. Culibrk, and D. Stefanovic, “Deep neural networks based recognition of plant diseases by leaf image classification,” vol. 2016, 2017.
 - [26] T. Rumpf, “Finding spectral features for the early identification of biotic stress in plants.” 2012.
 - [27] Y. Lanthier, A. Bannari, D. Haboudane, J. R. Miller, and N. Tremblay, “Hyperspectral Data Segmentation and Classification in Precision Agriculture : a Multi-Scale Analysis,” *Environment*. pp. 4–5.
 - [28] G. K. P. Vyshnavi, M. R. Sirpa, M. Chandramoorthy, and B. Padmapriya, “Healthy and Unhealthy Plant Leaf Identification and Classification Using Hierarchical Clustering.” pp. 448–453, 2016.
 - [29] S. Arivazhagan, R. N. Shebiah, S. Ananthi, and S. Vishnu Varthini, “Detection of unhealthy region of plant leaves and classification of plant leaf diseases using texture features,” *Agricultural Engineering International: CIGR Journal*, vol. 15, no. 1, pp. 211–217, 2013.
 - [30] A. S. Razavian, H. Azizpour, J. Sullivan, and S. Carlsson, “CNN features off-the-shelf: An astounding baseline for recognition,” *IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops*. pp. 512–519, 2014.
 - [31] D. S. Srdjan Sladojevic, Marko Arsenovic, Andras Anderla, “Deep Neural Networks Based Recognition of Plant Diseases by Leaf Image Classification,” *Comput. Intell. Neurosci.*, vol. 2016, 2016.
 - [32] C. Potena, D. Nardi, and A. Pretto, “Fast and Accurate Crop and Weed Identification with Summarized Train Sets for Precision Agriculture.” 2016.
 - [33] Z. Liu, J. Gao, G. Yang, H. Zhang, and Y. He, “Localization and Classification of Paddy Field Pests using a Saliency Map and Deep Convolutional Neural Network,” *Scientific Reports*, vol. 6, no. February. p. 20410, 2016.
 - [34] K. Simonyan and A. Zisserman, “Very Deep Convolutional Networks for Large-Scale Image Recognition,” *ImageNet Chall.*, pp. 1–10, 2014.
 - [35] S. Arlot and A. Celisse, “A survey of cross-validation procedures for model selection *,” *Statistics Surveys*, vol. 4, pp. 40–79, 2010.
 - [36] N. R. Cook, “Use and misuse of the receiver operating characteristic curve in risk prediction,” *Circulation*, vol. 115, no. 7, pp. 928–935, 2007.
 - [37] D. L. Streiner and J. Cairney, “What’s under the ROC? An introduction to receiver operating characteristics curves,” *Canadian Journal of Psychiatry*, vol. 52, no. 2, pp. 121–128, 2007.
 - [38] J. Schmidhuber, “deep learning in neural networks,” *Neural Networks*, vol. 61, pp. 85–117, 2015.
 - [39] L. Lu, H. Shin, H. R. Roth, M. Gao, L. Lu, S. Member, Z. Xu, I. Nogues, J. Yao, D. Mollura, and R. M. Summers, “Deep Convolutional Neural Networks for Computer-Aided Detection : CNN Architectures , Dataset Characteristics and Transfer Learning Deep Convolutional Neural Networks for Computer-Aided Detection : CNN Architectures , Dataset Characteristics and Transfer,” *IEEE Transactions on Medical Imaging*, vol. 35, no. 5, pp. 1285–1298, 2016.
 - [40] B. Athiwaratkun and K. Kang, “Feature Representation in Convolutional Neural Networks,” *arXiv:1507.02313 [cs]*. pp. 6–11, 2015.
 - [41] S. P. Mohanty, D. Hughes, and M. Salathé, “Using Deep Learning for Image-Based Plant Disease Detection,” vol. 7, no. September. pp. 1–7, 2016.
 - [42] A. K. Reyes, J. C. Caicedo, and J. E. Camargo, “Fine-tuning deep convolutional networks for plant recognition,” *CEUR Workshop Proceedings*, vol. 1391, 2015.